

Shinji Nishimoto, Ph.D.

Contact Info:

Graduate School of Frontier Biosciences, The University of Osaka
Yamadaoka 1-4, Suita, Osaka 565-0871, Japan
Email: nishimoto.shinji.fbs@osaka-u.ac.jp

Education:

2005 **Ph.D., Systems Neuroscience**

Graduate School of Engineering Science, Osaka University, Osaka, Japan

2002 **M.Eng., Systems Neuroscience**

Graduate School of Engineering Science, Osaka University, Osaka, Japan

2000 **B.S. (equivalent), Biophysical Engineering**

School of Engineering Science, Osaka University, Osaka, Japan

(Admitted to graduate school after three years of undergraduate study based on outstanding academic performance)

Academic experience:

2021-present, **Professor**

Institution: Graduate School of Frontier Biosciences, The University of Osaka, Japan

2013-2021, **Senior Researcher (Principal Investigator)**

Institution: National Institute of Information and Communications Technology, Japan

2005-2013, **Postdoctoral Fellow/Associate Specialist**

Institution: University of California, Berkeley, USA

Publications (past 5 years):

Matsuyama T, Nishimoto S*, Takagi Y* (2026) LaVCa: LLM-assisted Visual Cortex Captioning
International Conference on Learning Representations (ICLR) 2026 (*equal last author)

Yang H, Fukuma R, Namima T, Okuda K, Nishi A, Iwata T, Reza A, Sasaki KS, Kaiju T, Gill G, Kishima H,
Nishimoto S, Yanagisawa T. (2026) Longitudinal multitask wireless electrocorticography data from two fully
implanted nonhuman primates, *Scientific Data* 13: 63

Denk TI, Takagi Y, Matsuyama T, Agostinelli A, Nakai T, Frank C, Nishimoto S. (2026) Text-to-music generation
models capture musical semantic representations in the human brain, *Nature Communications* 17, 91

Asanuma H, Koide-Majima N, Nakamura K, Horii T, Nishimoto S, Oizumi M (2025) Correspondence of high
dimensional emotion structures elicited from video clips between humans and multimodal LLMs, *Scientific
Reports* 15: 32175

Yamashita M, Kubo R, [Nishimoto S](#) (2025) Conversational content is organized across multiple timescales in the brain, **Nature Human Behaviour** 9, 2066-2078

Nakai T, Kubo R, [Nishimoto S](#) (2025) Cortical representational geometry of diverse tasks reveals subject-specific and subject-invariant cognitive structures, **Communications Biology** 8,713

Wang J, Kawahata K, Blanc A, Maeda N, [Nishimoto S](#), Nishida S (2025) Asymmetric representation of symmetric semantic information in the human brain, **NeuroImage: Reports** 5: 100243

Nakagi Y, Matsuyama T, Koide-Majima N, Yamaguchi HQ, Kubo R, [Nishimoto S*](#), Takagi Y* (2024) Unveiling Multi-level and Multi-modal Semantic Representations in the Human Brain using Large Language Models, **Empirical Methods in Natural Language Processing (EMNLP)** 2024:20313-20338 (*equal last author)

Hayashi R, Kaji S, Matsumoto Y, Nishida S, [Nishimoto S](#), Takahashi H (2024) Homogenization of word relationships in schizophrenia: Topological analysis of cortical semantic representations, **Psychiatry and Clinical Neurosciences**, 78:687-695

Ikegawa Y, Fukuma R, Sugano H, Oshino S, Tani N, Tamura K, Iimura Y, Suzuki H, Yamamoto S, Fujita Y, [Nishimoto S](#), Kishima H, Yanagisawa T (2024) Text and image generation from intracranial electroencephalography using an embedding space for text and images, **Journal of Neural Engineering**, 21:036019

Okumura T, Kida I, Yokoi A, Nakai T, [Nishimoto S](#), Touhara K, Okamoto M (2024) Semantic context-dependent neural representations of odors in the human piriform cortex revealed by 7T MRI, **Human Brain Mapping**, 45:e26681

Koide-Majima N, [Nishimoto S](#), Majima K (2024) Mental image reconstruction from human brain activity: Neural decoding of mental imagery via deep neural network-based Bayesian estimation, **Neural Networks**, 170:349-363

Nakai T, [Nishimoto S](#) (2023) Artificial neural network modelling of the neural population code underlying mathematical operations, **NeuroImage**, 270:119980

Takagi Y, [Nishimoto S](#) (2023) High-resolution image reconstruction with latent diffusion models from human brain activity, **The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**, 2023: 14453-14463

Nakai T, [Nishimoto S](#) (2023) Quantitative modelling demonstrates format-invariant representations of mathematical problems in the brain, **European Journal of Neuroscience**, 57:1003-1017

Matsumoto Y, Nishida S, Hayashi R, Son S, Murakami A, Yoshikawa N, Ito H, Oishi N, Masuda N, Murai T, Friston K, [Nishimoto S](#), Takahashi H (2022) Disorganization of Semantic Brain Networks in Schizophrenia Revealed by fMRI, **Schizophrenia Bulletin**, sbac157

Nakai T, [Nishimoto S](#) (2022) Representations and decodability of diverse cognitive functions are preserved across the human cortex, cerebellum, and subcortex, **Communications Biology**, 5:1245

Chuyin Z, Koh ZH, Gallagher R, [Nishimoto S](#), Tsuchiya N (2022) What can we experience and report on a rapidly presented image? Intersubjective measures of specificity of freely reported contents of consciousness, **F1000Research**, 11:69

Cohen D, Nakai T, Nishimoto S (2022) Brain networks are decoupled from external stimuli during internal cognition, **NeuroImage**, 256: 119230

Fukuma R, Yanagisawa T, Nishimoto S, Sugano H, Tamura K, Yamamoto S, Iimura Y, Fujita Y, Oshino S, Tani N, Koide-Majima N, Kamitani Y, Kishima H. (2022) Voluntary control of semantic neural representations by imagery with conflicting visual stimulation, **Communications Biology**, 5: 214

Hatanaka G, Inagaki M, Takeuchi RF, Nishimoto S, Ikezoe K, Fujita I (2022) Processing of visual statistics of naturalistic videos in macaque visual areas V1 and V4, **Brain Structure and Function**, 227: 1385–1403

Nakai T, Koide-Majima N, Nishimoto S (2022) Music genre neuroimaging dataset, **Data in Brief**, 40: 107675

Konno D, Nishimoto S, Suzuki T, Ikegaya Y, Matsumoto N (2021) Multiple states in ongoing neural activity in the rat visual cortex, **PLoS One**, 16: e0256791

Shinkuma R, Nishida S, Maeda N, Kado M, Nishimoto S (2021) Reduction of Information Collection Cost for Inferring Brain Model Relations From Profile Information Using Machine Learning, **IEEE Trans. on Systems, Man, and Cybernetics: Systems**, 52: 4057-4068

Nishida S, Blanc A, Maeda N, Kado M, Nishimoto S (2021) Behavioral correlates of cortical semantic representations modeled by word vectors, **PLoS Computational Biology**, 17: e1009138

Nakai T, Yamaguchi HQ, Nishimoto S (2021) Convergence of Modality Invariance and Attention Selectivity in the Cortical Semantic Circuit, **Cerebral Cortex**, 31: 4825-4839

Ikutani Y, Kubo T, Nishida S, Hata H, Matsumoto K, Ikeda K, Nishimoto S (2021) Expert Programmers Have Fine-Tuned Cortical Representations of Source Code, **eNeuro**, 8: ENEURO.0405-20.2020

Nakai T, Koide-Majima N, Nishimoto S (2021) Correspondence of categorical and feature-based representations of music in the human brain, **Brain and Behavior**, 11: e01936

Book chapter (past 5 years):

Takagi Y, Nishimoto S (2024) Decoding models: From brain representation to machine interfaces, In Kay K (ed), *Computational and Network Modeling of Neuroimaging Data* (pp. 287-307), Academic Press.

Major Honor (past 5 years)

2023 JSPS Prize, Japan Society for the Promotion of Science
(One of Japan's most prestigious research awards)

Other Honors and Awards (past 5 years)

2025 Best Paper Award, Japanese Neural Network Society

2023 Distinguished Professor (honorary title), The University of Osaka

2021 3rd Place, Algonauts Challenge 2021 (Algonauts Project)

2021 Best Session Award, ISIS 2021 (International Symposium on Advanced Intelligent Systems)